System Overview

Our app is a personal finance tracking app which was designed to help everybody manage and understand their finances. Users will have full control of their finances by manually logging what comes in and out of their bank account with the reassurance of knowing that their bank account information does not have to be provided if that's what our users choose. Users will have the ability to categorize all their transactions, track monthly subscriptions, and have a visual dashboard to view weekly and monthly summaries easily. Built with our users' safety in mind our system will let them choose how they want to share their data while having full insight on how they are using their money all within our app.

From a user's perspective, our app will work like a journal but for personal finance. A user will be able to open up the app and immediately see the dashboard of their monthly expenses and recent transactions including their balance with a snapshot of how much they are adhering to their budget goals. From there they will be able to dive into different areas in our app where they can choose to log a new expense, check their recurring subscriptions with due dates, and even get help from an AI-Financier who will analyze spending patterns and generate tips based on that information.

Elements

- **User Account** will store the profile, preferences, and all financial data per user
- **Transaction log:** Recorded incomes and expenses, categories, date, amount
- **Budget Engine**: Comparing user spending with user defined saving goals
- **Subscription tracker**: A list of recurring charges reached marked with date, company, and amount per transaction
- **Savings:** where the user defines their spending limit goals
- **Spending overview**: Provided charts and visuals of monthly and weekly spending

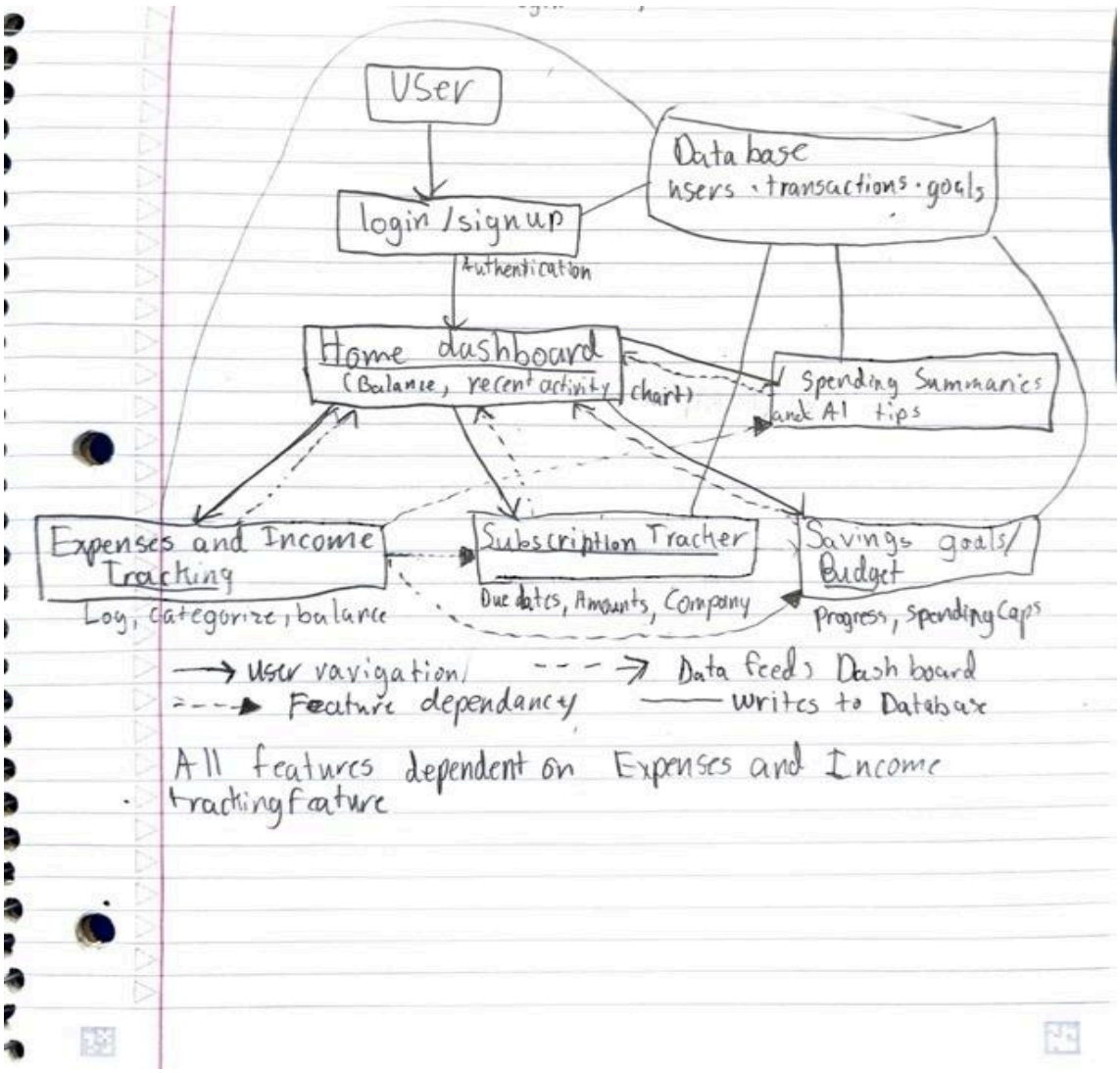
Relationships

- The transaction will feed data into the budget engine, savings limits, and spending summaries charts
- Subscriptions are classified as recurring transactions that have a schedule
- Savings goals are derived from the difference of income and expenses

Behaviors

- **User logs transactions ->** System will update the balance, update savings progress, visuals will get updated as well
- **Subscription payment date is near ->** System will send reminders whether via email or notify on phone
- **User sets a spending/savings goal** -> System will auto track progress the goal through calculations from income and spending as they are logged

System Sketch



Rationale

Our main goal with creating this app is for people to become more financially literate and aware about their spending or budgeting. They would be able to do it all in one place without having to give their credit card or bank account numbers if they don't want to. One issue that we have observed with using other apps is that there is a requirement for giving your bank account and credit card numbers in order to use their service. In our case, if the user chooses not to give their accounts it would result in more work for them (making sure they upload every single purchase, inputting their paychecks, etc.), but they wouldn't have to give up sensitive information if they don't want to.

The few big features in our application will be tracking savings and savings goals, customizable categories, monthly and weekly summaries, and subscription overviews and reminders. Since we are building a product to help with financial literacy, keeping track of how much a person has in savings will help towards that and if the user has a particular amount they want to save up to, it will also track their progress in their savings goals. Inputting customized categories will allow users to organize their purchases in a way that makes sense to them, which encourages them to think about what purchases truly matter. Monthly and weekly outlooks allow the users to see what their money is going into on a regular basis, and they can get a better idea of how much they spend in a given month or week based on surrounding circumstances. In the same vein, subscription overviews would show users which subscriptions they have active and how much each one costs. From the overview, they can have a better view of which of their subscriptions they believe would be worth keeping instead of having multiple subscriptions floating around that they never use.

Methodology

Our team gathered requirements for the system by at first identifying the main problem our idea is trying to solve, which is to help users manage and track their finances. To see what users want from a budgeting app, we started researching existing apps that do similar functions to ours and looked at how they allow users to track expenses and display spending summaries or charts. We then analyzed the ideas we came up with to determine which ones would be realistic with deadlines and the most useful ones. While we were doing this, we talked about how each feature would affect the user and how it would help them solve the main issue and if it could actually be implemented in the timeline that we have for the project. To validate the requirements, we reviewed the features as a team and made them smaller and better by using user stories. Writing user stories helped us think about how a user would interact with the system and what they would gain from using this feature.

When coming up with the idea, we talked about ideas based on personal experiences with budgeting and tracking costs. This helped come up with potential features and the most useful features for what we would use a budgeting app for. To evaluate requirements, we looked at if the feature helps the user better track and manage their money, and whether the features could be done in time for the project deadlines. We also made sure that the requirements are clearly defined and address users' needs without being overly complex.

We use FDD to organize our ideas around features that will provide beneficial value to users, like tracking expenses, or spending summaries. XP splits features into smaller tasks that can be implemented step by step. SCRUM helps us organize our development and will help the team track progress and complete tasks.

Our team is focusing on features because they represent what the user can accomplish with our app. User stories help make the features better by viewing them from the customers perspective and looking at the values they provide. This project is in 2 week sprints, our team is meeting once a week on Friday where we delegate tasks to the assignment and work on then if time and then update on work that had been done since the last meeting. And these meetings will end up with a MVP by the end of every 2 weeks.

Requirements Engineering

1. Monthly/weekly spending overview
    a. **Gaby:** The monthly and weekly spending overview feature is important to have for the user because the user would like to see how much they are spending. Either monthly or weekly, this will allow the user to make financial decisions based on the spending that occurs monthly and weekly. By displaying this to the user, they would be able to see visually. The reason why we think it would be a good feature to have for the app is because the user should be able to see how they are spending their money whether it is good or bad. Just having an overview display of monthly and weekly overview will help the user understand where their money is going.This feature will be very helpful for users who are trying to become more aware about how their money is being spent. Helping them understand, whether they want to see where they spend most money or how little they are spending money, they want to see that and how it affects their financial goals. For a user interaction system and system interface behavior, for this specific feature there wouldn't be any input that is needed because this feature is just showing the spending that is being done monthly and weekly. Which is deriving from other parts of the app to create this overview. Overall the experience that the user would get from this would allow them to see the bigger picture about the way they are spending their money.
    b. **Madison:** A user earns a steady income but does not have a way to track their spending. At the end of the month the user looks at their bank account and is stressed about where all the money they thought they had went. After using our app, they open it up and see the monthly overview. It shows that they spent 50% of their income on fast food and going out to eat. They can then switch to the weekly spending tab and see that they spent over 100 dollars on food that week alone. Without doing any math or going through past bank statements, the user only has to input the money they spend in the system. The user will then easily see a pattern in their spending and see their habits. The user does not need to be

a financial expert or have a spreadsheet. Our app shows them a month or a week of transactions in one spot to make the user more aware of their spending habits all in one spot. The app replaced that stress of wondering where all the money went to a clear representation of what they spent their money on.

c. **Walter:** A user before using our app would get paid on friday but would have no idea where the money had gone by next monday. They knew they had many bills to pay but would be surprised when the small purchases added up which for example would be coffee, gas, and food. These purchases would drain the users bank account silently and cause them to not be able to pay their bills on time. After using our product they were able to easily track their expenses by looking at the weekly spending overview, set budget goals, and had reminders set on upcoming bills that they need to pay. The weekly view helped them by making sure they weren't over spending during the week and not be faced with surprises before it was too late.

d. **Matt:** A user would want a clear way to view their income expenses, this can be done through weekly and monthly overviews. Users would be able to use this information to understand trends in their spending patterns, identify unnecessary expenses, and make more informed budgeting decisions without needing to connect a bank account or rely on other external tools. Not only would this give the user the ability to see trends in their financial behavior and recognize whether they're staying within a given budget, but it would also give the user a sense of control over their financial activity.

e. **Shafiya:** Without a way to keep track of their regular spending, users wouldn't know how much money they have spent. If they are able to see a number, they will know if they spent too much this week and need to cut back for next week or if they went under budget and they can afford to indulge in more things the following week.

2. Track savings
    a. **Gaby**: The purpose of this feature is to help the user track how much savings they have. Whether they have more or less on a given date or if the user wants to track it to get to their savings goal. This feature is helpful to give users a clear idea about the finances of their savings. For user interaction system and system interface behavior, there would be a place on the display where the user would have to input how much is either going to their savings or getting deducted. For the user to let the app know that they are either adding or deducting, the user would be inputting the amount next to the amount they have in savings, and there would be a toggle bar where there will be two options (adding or deducting). The user would have to select one of them to let the app know so the app is able to make the right calculation. Then they would be able to see their previous inquiries with either adding or deducting from their savings. If the user has set a goal for their savings, then there would be a display that would show how much more is needed till they are at their goal.
3. Subscription overview
    a. **Gaby**: This feature would be users that want to be able to keep track of all the subscriptions that they have currently connected to their accounts. With this overview, they will see a display with all the current active subscriptions that they have, as well as when the next due date is for each subscription and how much it is so the user is aware of the charges that they are to get. There will be a section on the display to let the app know when they have cancelled a subscription so the app will be able to take it off from the current active list of subscriptions. The user will also be able to see any previous inactive subscription that they had. For user system interaction and system interface behavior, the parts where the user would be interacting with the app is to update whether the charge went through and if that subscription was paid. They will also be able to let the app know which subscription is becoming inactive so the app is able to put it in the inactive subscription overview.

4. Categorize spending
    a. **Gaby**: This feature is very helpful for users to have because they will now be able to put categories of where their spending is going. Let's take an example case. A user has put into the area where the app asks how much money they would like to enter and with the date. After doing so, the user will select the user-made categories (savings, groceries, gas, shopping,etc.) that they would like to put to categorize the transaction they have done. The app is giving the user the ability to create the categories because every user is different and wants to organize differently. For user system interaction and system interface behavior, there will be an area on the display to prompt the user to input a valid number to represent the money spent in that transaction. Then the user would be prompted to input the date in mm/dd/yyyy order. After, there will be a toggle dropdown that will show all the options that the user created and with an option to create a new category if the user wanted to add another category to the list of categories.

Minimum Viable Product

Github: https://github.com/gabech-1908/CS-321-Project.git

To develop a Minimum Viable Product (MVP) for this system, we first have to ask ourselves what our core feature is. As stated in the rationale, our main goal with creating this app is for people to become more financially literate and aware about their spending or budgeting by giving users full control over their financial data without requiring sensitive information.

User Stories:

1. Manual Income Tracking
    - *As a user, I would want to be able to input income manually so as to stay aware of their earnings.*
2. Manual Expense Tracking
    - *As a user, I would want to log purchases to understand where my money is going.*
3. Spending Awareness (Balance + Summaries)
    - *As a user, I want to be able to view how much I've spent vs earned so I can manage my budget.*
    - *As a user, I want to identify my spending patterns through weekly/monthly summaries.*
4. Subscription Awareness
    - *As a user, I want to be aware of and track recurring expenses.*

Features + Functions:

1. Income Management
    - Input income amount
    - Store income value
    - Update/edit income
2. Expense Management

    - Add expense (display amount, category, date)
    - Store list of expenses
    - Delete/edit expense
3. Budget Overview
    - Calculate total expenses
    - Compute remaining balance:

```
balance = income - total_expenses;
```

    - Display summary (income, expenses, balance)

Task Breakdown

Backend / Logic Tasks

- Create **Income** variable/storage
- Create **Expense** class/object:
    - amount
    - category
    - date
- Implement expense list (array/list)
- Write function to compute total expenses
- Write function to compute balance
- Write function to compute weekly summary
- Write function to compute monthly summary

Frontend Tasks

- Build income input form
- Build expense output form
- Display expense list
- Display weekly/monthly summary (income, expenses, balance)

Integration Tasks

- Connect UI inputs to data model
- Update display dynamically after input

- Validate inputs (no negative values, etc.)